

Programming for Essential Digital Skills, Chapter 9

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Pyplot

- The most common way to plot in Python is to use the `pyplot` module contained in the library `matplotlib`.
- By convention we load it as `plt`:

```
import matplotlib.pyplot as plt
```

House Price Data as an Example

- To illustrate the plotting methods, we will use the house price data from the previous chapter as an example.
- We repeat the cleaning steps:

```
import pandas as pd
df = pd.read_csv("HousingPrices-Amsterdam-August-2021.csv")
df.drop('Unnamed: 0', axis=1, inplace=True)
df = df[~df['Price'].isna()]
df['price_per_sqm'] = df['Price'] / df['Area']
df.rename(columns={"Zip" : "postcode", "Room" : "num_rooms"},
          inplace=True)
df.columns = df.columns.str.lower()
```

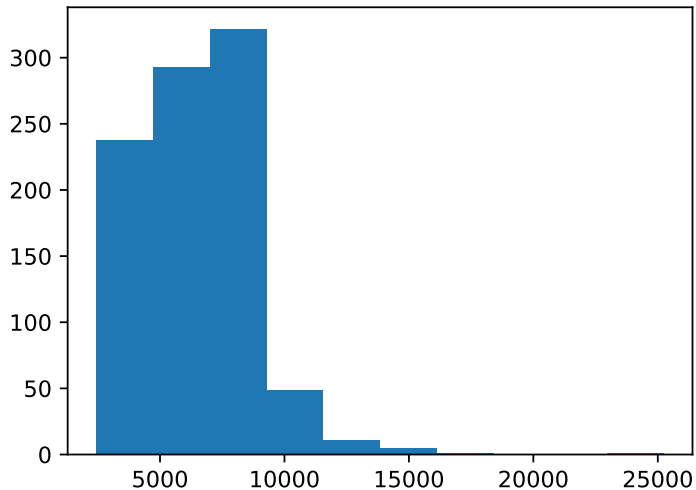
Histograms

- To describe the distribution of a single numeric variable, we can use a histogram.
- A histogram splits the data into “bins” and shows the number of observations in each bin.

```
plt.hist(df["price_per_sqm"])
```

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Histograms



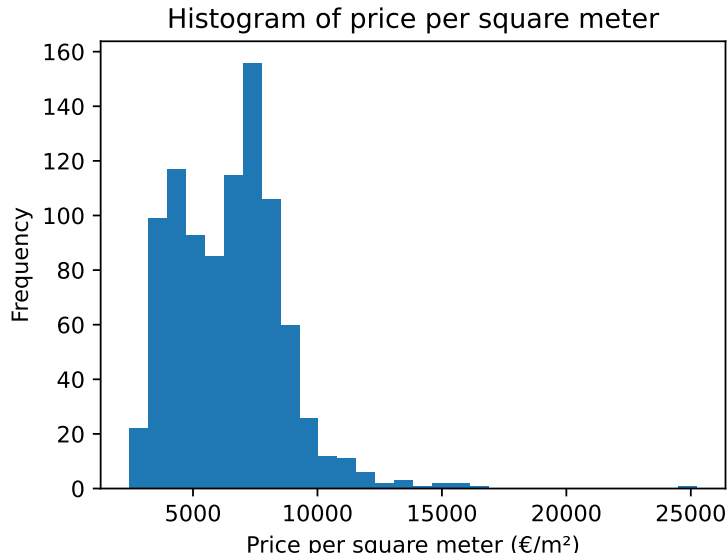
Histograms

- We can control the number of bins and adjust axis labels as follows:

```
plt.hist(df["price_per_sqm"], bins=30)
plt.xlabel("Price per square meter (€/m²)")
plt.ylabel("Frequency")
plt.title("Histogram of price per square meter")
```

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Histograms



Bar plot

- A bar plot is useful when we want to visualize the number of occurrences of a variable.
- This can be both for numeric and categorical variables.
- For example, suppose we want to visualize the output of:

```
df["num_rooms"].value_counts()
```

```
num_rooms
3      330
4      201
2      191
5       97
6       42
7       19
1       17
8       11
9        6
```

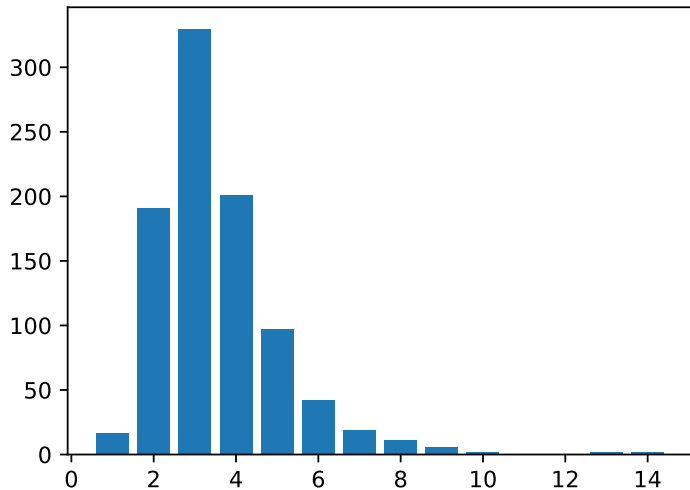
Bar plot

We can do that with:

```
num_room_counts = df["num_rooms"].value_counts()  
plt.bar(num_room_counts.index, num_room_counts.values)
```

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Bar plot



Scatter Plot

- Histograms and bar plots are used to describe a single variable.
- When we want to describe the relationship between a pair of variables we use a scatter plot.
- This shows all the values of the two variables on a Cartesian plane.

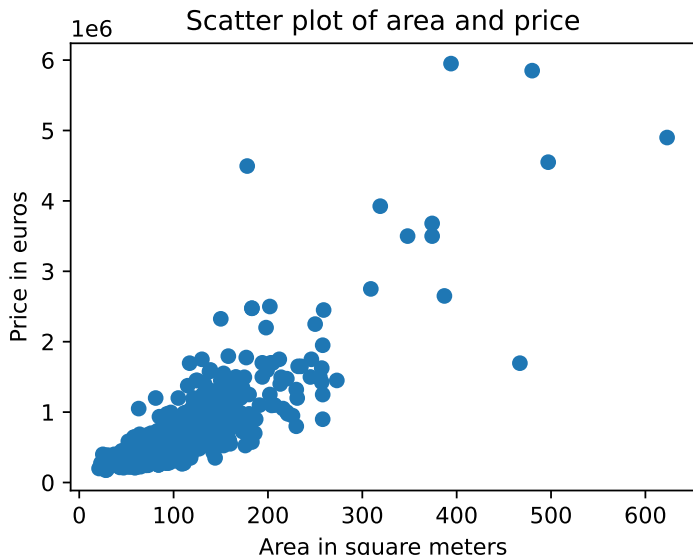
Scatter Plot

- We show how price and area move together:

```
plt.scatter(df["area"], df["price"])  
plt.xlabel("Area in square meters")  
plt.ylabel("Price in euros")  
plt.title("Scatter plot of area and price")
```

(Output on next slide)

Scatter Plot



Scatter Plot

- The vertical axis is scaled with $1e6$, which is less easy to read.
- We can scale the price variable by 1,000,000 and change the variable label accordingly:

```
plt.scatter(df["area"], df["price"] / 1000000)
plt.xlabel("Area in square meters")
plt.ylabel("Price in millions of euros")
plt.title("Scatter plot of area and price")
```

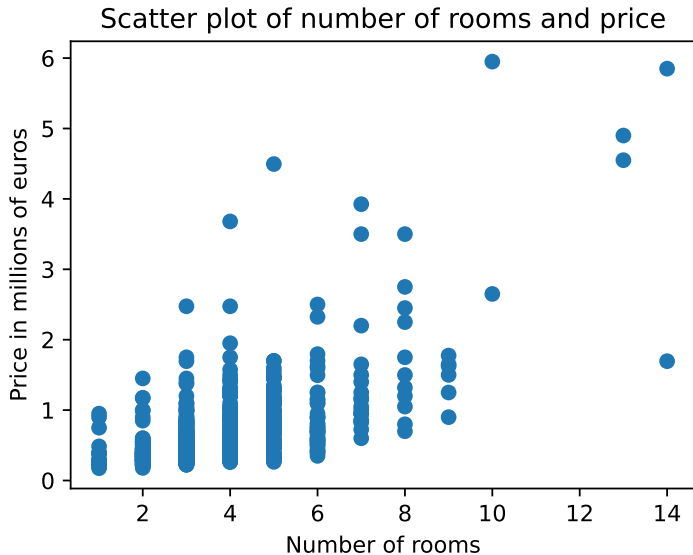
Plotting Average by Group

- Suppose we plot price against the number of rooms:

```
plt.scatter(df["num_rooms"], df["price"] / 1000000)
plt.xlabel("Number of rooms")
plt.ylabel("Price in millions of euros")
plt.title("Scatter plot of number of rooms and price")
```

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Plotting Average by Group



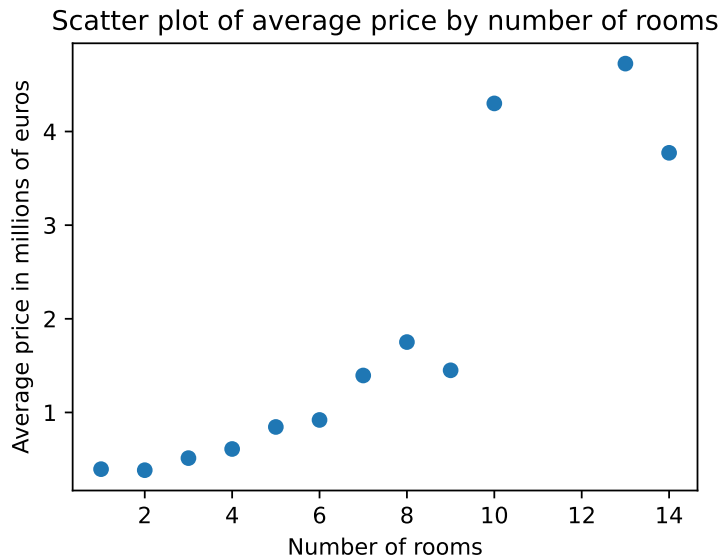
Plotting Average by Group

- In some cases (like this one), it can be more useful to plot the average of a variable by the value of another, or bins of values.
- We can do that as follows:

```
plot_df = df.groupby("num_rooms", as_index=False)["price"].mean()  
plt.scatter(plot_df["num_rooms"], plot_df["price"] / 1000000)  
plt.xlabel("Number of rooms")  
plt.ylabel("Average price in millions of euros")  
plt.title("Scatter plot of average price by number of rooms")
```

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Plotting Average by Group



Using Colors to Visualize 3 Variables

- We can also use colors to visualize 3 variables at once.
- The color of the points can represent the values of a 3rd variable.
- We can do this as follows:

```
plt.scatter(df["area"], df["price"] / 1000000, c=df["num_rooms"])  
plt.xlabel("Area in square meters")  
plt.ylabel("Price in millions of euros")  
plt.colorbar(label="Number of rooms");
```

(Output on next slide)

Using Colors to Visualize 3 Variables

